Lab 3: Genetics of *Drosophila melanogaster*

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Abstract

The purpose of this experiment was to identify and analyse genetic linkages in *Drosophila melanogaster* by crossing individuals expressing mutant traits with wild type mates. Assuming Mendel's Laws hold true, we identified females with five visible mutant traits and crossed them with wild type males. We collected their progeny (F1) and then analyzed their phenotypes to see if certain genes were on the same chromosome, as well as sex linkage.

Introduction

Gregor Mendel, also known as the father of genetics worked extensively with pea plants. Through his experiments, he observed that the plants had key discrete traits, and these traits were passed on from parents to offspring. These traits could present themselves in various forms, and these variants came to be known as alleles. As a result of his observations, he proposed three laws of inheritance that govern the transmission of these heritable characteristics from parents to offspring.

1. Law of Segregation: Alleles for a trait separate during gamete formation, and each gamete receives only one allele.
2. Law of Independent Assortment: Alleles for different traits assort independently of each other during gamete formation.
3. Law of Dominance: In a heterozygote, the dominant allele masks the expression of the recessive allele.

During Mendel's time however, chromosomes had not yet been identified as the carriers of heredity. So, the concept that different traits could be passed down together, did not exist. It was later discovered by Thomas Hunt Morgan, that genes are located on chromosomes, and that the closer two genes are on a chromosome, the less likely they are to be separated during gamete formation.

In modern *Drosophila* genetics, researchers often use **balancer chromosomes** to manage complex genotypes and study linkage. Balancer chromosomes are engineered with multiple chromosomal inversions that effectively suppress recombination between homologous chromosomes during meiosis. They typically carry a dominant marker phenotype, such as Curly wings (Cy), allowing researchers to track the inheritance of specific chromosomes across generations and maintain lethal or sterile mutations without the risk of them being lost to crossing over. Generally, these mutuations are dominant, and are lethal when inheritance homozygously. By ensuring that the balanced chromosome segregates as a single, discrete unit with its markers, researchers can more reliably apply Mendel's laws of segregation and independent assortment to identify linkage and map genetic traits.

In our flies, the female parents had the genotype $\frac{w,y}{+};\frac{Cy}{Scu};\frac{Stu}{+}$ and the wild type male parents had the genotype $\frac{+}{Y};\frac{+}{+};\frac{+}{+}$. We know that the females contained the CyO Balancer Chromosome.

Materials and Methods

Materials

The female flies we used contained the following mutations: white eyes, yellow body, scutoid, stubble, curly wings. They were mated with wild type males with the following traits: red eyes, brown body, no scutoid, normal hair, straight wings. They were fed with cornstarch feed, and some yeast paste was applied to promote egg laying. The flies were handled on a flypad, and anaesthetized using CO_2 gas. A fine brush was used to handle them, and they were observed under a dissection stereomicroscope.

Methods

In a cornstarch food vial, we applied a small amount of yeast paste to promote egg laying. We then added 6 female flies with the markers and placed them in the vial along with 3 wild type males. We left the vials for 14 days to allow the flies to mate and lay eggs. On the 14th day, we collected the F1 progeny, anaesthetized them and recorded the number of male and female flies with each phenotype.

Results

F1 Progeny Phenotypes

Gender	Eye Colour	Wing Shape	Scutoid	Stubble	Body Colour
Female	Red	Curly	No	No	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Curly	No	Yes	Brown
Female	Red	Straight	Yes	No	Brown
Female	Red	Straight	Yes	No	Brown
Female	Red	Straight	Yes	No	Brown
Female	Red	Straight	Yes	No	Brown
Female	Red	Straight	Yes	No	Brown
Female	Red	Straight	Yes	No	Brown
Female	Red	Straight	Yes	Yes	Brown
Female	Red	Straight	Yes	Yes	Brown
Female	Red	Straight	Yes	Yes	Brown
Female	Red	Straight	Yes	Yes	Brown
Female	Red	Straight	Yes	Yes	Brown
Female	Red	Straight	Yes	Yes	Brown
Female	Red	Straight	Yes	Yes	Brown
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Male	White	Curly	No	Yes	Yellow
Male	White	Curly	No	Yes	Yellow
Male	White	Curly	No	Yes	Yellow
Male	White	Curly	No	Yes	Yellow
Male	White	Straight	No	No	Yellow
Male	White	Straight	Yes	No	Yellow
Male	White	Straight	Yes	No	Yellow
Male	White	Straight	Yes	No	Yellow
Male	White	Straight	Yes	No	Yellow
Male	White	Straight	Yes	Yes	Yellow
Male	White	Straight	Yes	Yes	Yellow
Male	White	Straight	Yes	Yes	Yellow
Male	White	Straight	Yes	Yes	Yellow
Male	White	Straight	Yes	Yes	Yellow
Male	White	Straight	Yes	Yes	Yellow

Discussion

Sex Linkage

Eye colour and body colour seem to be sex-linked: Out of 22 females, all of them have red eyes and brown bodies, whereas the remaining 18 males have white eyes and yellow bodies.

Autosomal Chromosome Mapping

1. There seems to be a strong correlation between the wing shape and scutoid phenotypes: 15 flies with curly wings have no scutoid, and 24 flies have straight wings and scutoid. There is only 1 outlier with straight wings and scutoid, which is possibly observational error. This suggests that the genes for wing shape and scutoid are located on the same chromosome, with the dominant alleles for both on separate homologous chromosomes.
2. The distribution of stubble or no stubble does not seem to correlate with any other visible markers according to the data, so it probably is on a separate chromosome with respect to the other markers

Dominant and Recessive Alleles

Eye and body colour

The female parents had white eyes and yellow bodies (w, y), and the wild type male parents had red eyes and brown bodies $(+)$.

Case 1: Female genotype: $\frac{w,y}{+}$, Male genotype: $\frac{+}{Y}$

In this case 50% of the male progeny would have the genotype $\frac{w,y}{Y}$ and the remaining 50% would have the genotype $\frac{+}{Y}$. Similarly, 50% of the female progeny would have the genotype $\frac{w,y}{+}$ and the remaining 50% would have the genotype $\frac{+}{+}$.

	$+$	$\mathbf{Y}$
$\mathbf{w, y}$	$\frac{w,y}{+}$	$\frac{w,y}{Y}$
$+$	$\frac{+}{+}$	$\frac{+}{Y}$

Case 2: Female genotype: $\frac{w,y}{w,y}$, Male genotype: $\frac{+}{Y}$

In this case, 100% of the male progeny would have the genotype $\frac{w,y}{Y}$ and 100% of the female progeny would have the genotype $\frac{w,y}{+}$.

	$+$	**Y**
w, y	$\frac{w,y}{+}$	$\frac{w,y}{Y}$
w, y	$\frac{w,y}{+}$	$\frac{w,y}{Y}$

From our results, 18 out of 40 flies have white eyes and yellow bodies (all males), and the remaining 22 out of 40 flies have red eyes and brown bodies (all females). If Case 1 had been true, we would expect close to 50% of the F1 males to have red eyes and brown bodies, but this was not the case. So, Case 2 must be true, meaning that the wild type genes for eye and body colour are dominant over the mutant genes, and these genes are located on the X chromosome.

Wing Shape

The female parents had curly wings (Cy), and the wild type male parents had straight wings ($+$).

Case 1: Female genotype: $\frac{Cy}{+}$, Male genotype: $\frac{+}{+}$

In this case 50% of the total progeny would be $\frac{Cy}{+}$, and the remaining 50% would be $\frac{+}{+}$.

	$+$	$+$
Cy	$\frac{Cy}{+}$	$\frac{Cy}{+}$
$+$	$\frac{+}{+}$	$\frac{+}{+}$

Case 2: Female genotype: $\frac{Cy}{Cy}$, Male genotype: $\frac{+}{+}$

In this case, 100% of the progeny would be $\frac{Cy}{+}$.

	$+$	$+$
Cy	$\frac{Cy}{+}$	$\frac{Cy}{+}$
Cy	$\frac{Cy}{+}$	$\frac{Cy}{+}$

From our results, 15 out of 40 flies are curly winged, and the remaining 25 are straight winged. So, 37.5% of the progeny are curly winged. If Case 2 had been true, we would expect 100% of the progeny to show the same phenotype, which is not what we observed. So, Case 1 must be true, meaning that the curly wing gene is dominant over the mutant gene.

Scutoid

The female parents had scutoid (Scu), and the wild type male parents had no scutoid (+).

Case 1: Female genotype: $\frac{Scu}{+}$, Male genotype: $\frac{+}{+}$

In this case 50% of the total progeny would be $\frac{Scu}{+}$, and the remaining 50% would be $\frac{+}{+}$.

	+	+
Scu	$\frac{Scu}{+}$	$\frac{Scu}{+}$
+	$\frac{+}{+}$	$\frac{+}{+}$

Case 2: Female genotype: $\frac{Scu}{Scu}$, Male genotype: $\frac{+}{+}$

In this case, 100% of the progeny would be $\frac{Scu}{+}$.

	+	+
Scu	$\frac{Scu}{+}$	$\frac{Scu}{+}$
Scu	$\frac{Scu}{+}$	$\frac{Scu}{+}$

From our results, 24 out of 40 flies are scutoid. So, 60% of the progeny are scutoid. If Case 2 had been true, we would expect 100% of the progeny to show the same phenotype, which is not what we observed. So, Case 1 must be true, meaning that the scutoid gene is dominant over the mutant gene.

Stubble

The female parents had stubble (Stu), and the wild type male parents had no stubble (+).

Case 1: Female genotype: $\frac{Stu}{+}$, Male genotype: $\frac{+}{+}$

In this case 50% of the total progeny would be $\frac{Stu}{+}$, and the remaining 50% would be $\frac{+}{+}$.

	+	+
Stu	$\frac{Stu}{+}$	$\frac{Stu}{+}$
+	$\frac{+}{+}$	$\frac{+}{+}$

Case 2: Female genotype: $\frac{Stu}{Stu}$, Male genotype: $\frac{+}{+}$

In this case, 100% of the progeny would be $\frac{Stu}{+}$.

	+	+
Stu	$\frac{Stu}{+}$	$\frac{Stu}{+}$
Stu	$\frac{Stu}{+}$	$\frac{Stu}{+}$

From our results, 25 out of 40 flies have stubble. So, 62.5% of the progeny are stubbled. If Case 2 had been true, we would expect 100% of the progeny to show the same phenotype, which is not what we observed. So, Case 1 must be true, meaning that the stubble gene is dominant over the mutant gene.

References

1. Gregor Mendel and the Principles of Inheritance
2. Introduction to Balancer Chromosomes — Bloomington Drosophila Stock Center